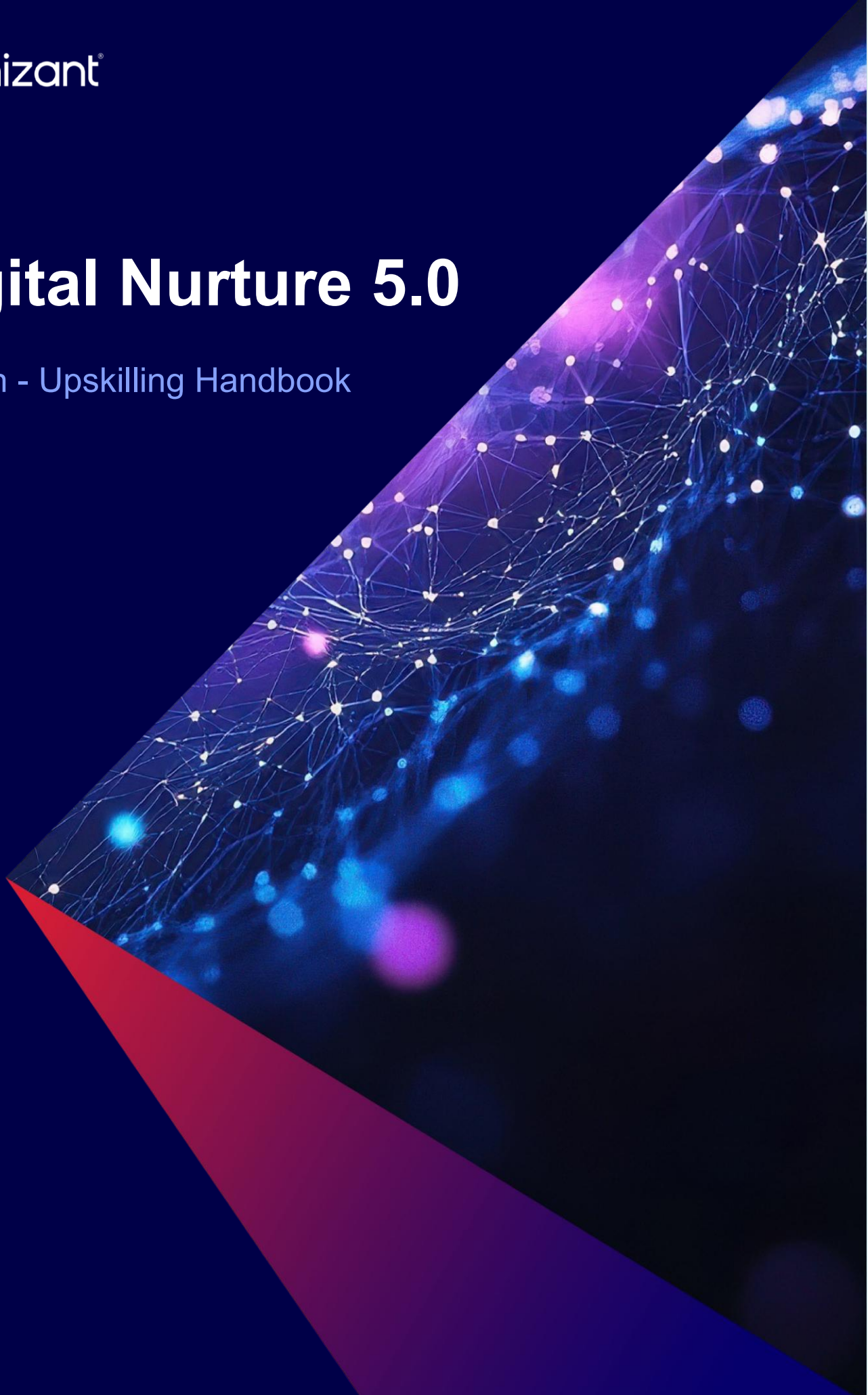




Digital Nurture 5.0

Python - Upskilling Handbook

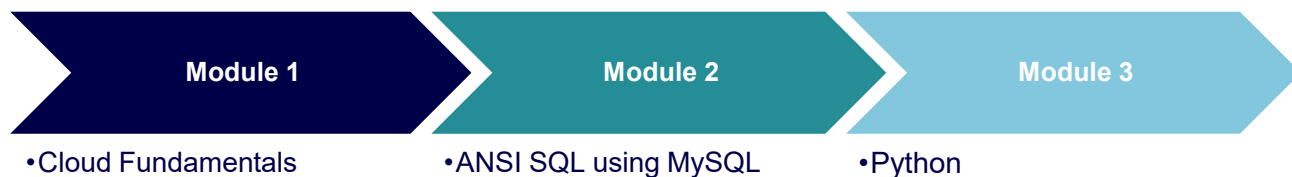


Program Highlights

- The Upskilling learning program has 3 Modules to be completed. Go through the recommended self- learning resources and practice the exercises to excel in the recommended Python FSE modules.
- Upskilling covers the below mentioned skills.
 - Cloud Fundamentals | ANSI SQL using MySQL | Python
- Please aim to complete the learning in 3 weeks time.

Recommended Program Sequence

The learning journey contains the following modules, followed by an Upskilling Final Assessment.



Learning Approach

Self-paced learning through open-source learning reference links



Self-Paced Learning using Open-Source Reference Links

- Please refer to the learning reference links provided to learn and understand the recommended concepts.
- We expect you to dedicate **2-3 hours of focused attention daily** and **schedule longer sessions on weekends** for your learning modules.
- Balance your learning with regular academic commitments.

Disclaimer: Cognizant does not claim ownership or responsibility for the content or any issues with the links provided, as they are merely references available on the internet. Candidates are free to leverage additional sources beyond what has been provided to enhance their skill capabilities for upskilling.

Effective learning strategies

Create a Study Schedule	Set Clear Goals	Stay Organized	Active Learning
Dedicate specific times daily for learning.	- Define what you want to achieve. - Break down tasks into manageable chunks.	- Keep track of your progress and deadlines. - Use tools like calendars, to-do lists, and reminders.	- Engage with the material through hands-on exercises. - Practice coding and solving problems regularly. - Ensure to complete all exercises.

Online Code Editor

Online Code Editors	URL
SQL	https://www.mycompiler.io/new/sql
	https://www.jdoodle.com/execute-sql-online/
	https://www.tutorialspoint.com/execute_sql_online.php
	https://www.w3schools.com/sql/trysql.asp?filename=trysql_asc
	https://onecompiler.com/mysql
Python	https://www.tutorialspoint.com/compilers/online-python-compiler.htm
	https://www.onlinegdb.com/online_python_compiler
	https://www.jdoodle.com/python-programming-online
	https://www.w3schools.com/python/python_compiler.asp
- Set up Integrated Development Environments (IDEs) like JupyterLab, Visual Studio Code (VS Code) , Spyder or PyCharm. - Install necessary tools and libraries as per course requirements.	

Exercise Instructions

In this learning program, you will be required to complete exercises designed to reinforce the concepts learned. These exercises are hosted on a public GitHub repository and must be downloaded and solved. Follow the instructions below to ensure a smooth and productive exercise workflow:

1. Access the Exercises:

- Exercises will be made available in our public GitHub repository.
- The repository URL is <https://github.com/seshadrimr/Digital-Nurture-Python>
- Navigate to the repository and locate the folder named **Python -> Upskilling**. Inside it, you will find a set of exercises.

2. Download the Exercises:

- Download the files for the exercises by clicking the download button on the GitHub repository page.

3. Solving the Exercises:

- Solve the problem statements provided in the downloaded files.
- Ensure that you understand the problem requirements and apply the concepts learned.
- Take your time to think through the solutions and code them accurately.

4. Self-Evaluation:

- After completing the exercises, evaluate your solutions based on the problem criteria.
- Compare your approach with any hints or solutions provided if available.
- Reflect on any mistakes or areas where you can improve.

5. Submitting Solutions:

- Firstly, organize your solutions and keep them in a folder.
- Create a **public repository** in your personal GitHub account, upload your solution folder and share the URL with the POC **on demand**.

6. Additional Support:

- If you encounter difficulties or have questions about the exercises, seek help from peers.
- Utilize the resources and links provided in this handbook for further assistance.

Module 1

Cloud Fundamentals

Overview:

This module provides a strong foundation in Cloud Computing by introducing learners to the fundamental concepts that drive modern IT infrastructure. Participants will explore the basics of cloud computing, including cloud service models (IaaS, PaaS, and SaaS), cloud deployment models, and the role of virtualization in enabling scalable and flexible computing environments.

The module also familiarizes learners with major cloud providers and basic cloud services such as compute, storage, and networking. Through real-world examples and conceptual exercises, participants will gain a clear understanding of how cloud technologies are used to deliver reliable, cost-effective, and on-demand IT services, preparing them for further learning and practical application in cloud-based environments.

Learning Objectives:

By the end of this module, learners will be able to:

- Understand the fundamentals and benefits of Cloud Computing
- Explain different Cloud Service Models (IaaS, PaaS, SaaS)
- Identify various Cloud Deployment Models
- Understand the concept and role of Virtualization in cloud
- Recognize major Cloud Service Providers
- Understand basic Cloud Services such as Compute, Storage, and Networking

Self-Learning (Open-source links):

Cloud Fundamentals		
Topics	Sub-Topics	Learning Reference Links
Introduction	Definition of Cloud Computing, Need for Cloud, Benefits, Characteristics of Cloud	https://www.geeksforgeeks.org/cloud-computing/
Cloud Service Models	Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Comparison of Service Models	https://www.geeksforgeeks.org/software-engineering/difference-between-iaas-paas-and-saas/
Cloud Deployment Models	Public Cloud, Private Cloud, Hybrid Cloud, Community Cloud, Use Cases	https://www.geeksforgeeks.org/cloud-deployment-models/
Virtualization	Virtual Machines, Hypervisors (Type 1 & Type 2), Benefits of Virtualization, Virtualization vs Cloud	https://www.geeksforgeeks.org/cloud-computing/virtualization-cloud-computing-types/
Cloud Providers	Overview of AWS, Microsoft Azure, Google Cloud Platform (GCP), Comparison of Providers	https://www.geeksforgeeks.org/blogs/top-cloud-platform-service-providers/
Basic Cloud Services	Compute Services, Storage Services, Networking Services (Conceptual Overview)	https://www.geeksforgeeks.org/cloud-computing/cloud-computing/

Check Your Understanding:

[Cloud Fundamentals Quiz](#)

ANSI SQL using MySQL

Overview:

This module provides a comprehensive introduction to ANSI SQL standards with practical application using MySQL. It covers the importance of SQL standardization, basic to advanced querying techniques, data manipulation operations, and schema management. Learners will start by understanding the foundation of SQL and gradually move into querying data with SELECT statements, applying filters, sorting results, using aggregate functions, and performing table joins. The module further explores working with subqueries, manipulating data within tables, and managing database schema objects. It also focuses on ensuring performance and integrity through indexes and constraints. Upon completion, learners will have a strong foundational knowledge to work with relational databases using SQL confidently and efficiently.

Learning Objectives:

By the end of this module, learners will be able to:

- Understand the role and significance of ANSI SQL and MySQL in relational database management.
- Retrieve and manipulate data from relational databases using standard SQL queries.
- Apply filtering, sorting, and aggregation techniques to manage query outputs effectively.
- Implement advanced querying techniques using joins and subqueries to combine and analyze complex data sets.
- Perform data manipulation operations including inserting, updating, and deleting records.
- Design and manage database structures by creating, altering, and dropping tables.
- Enhance database performance and enforce data integrity using indexes and constraints.

Self-Learning (Open-source links):

ANSI SQL using MySQL		
Topics	Sub-Topics	Learning Reference Links
Introduction to ANSI SQL and MySQL	Overview of ANSI SQL, Introduction to MySQL, Importance of Standard SQL	https://www.dwbi.org/pages/30/introduction-to-ansi-sql
		https://www.w3schools.com/mysql/mysql_intro.asp
Data Retrieval with SELECT Statement	Basic SELECT syntax, Retrieving specific columns, Filtering data with WHERE clause, Sorting results with ORDER BY	https://www.geeksforgeeks.org/mysql-select-statement/
Filtering and Sorting Data	Using logical operators in WHERE clause, Combining conditions with AND, OR, Sorting data using multiple columns	https://www.sqlshack.com/learn-mysql-sorting-and-filtering-data-in-a-table/
Aggregate Functions and Grouping	Introduction to aggregate functions (SUM, AVG, COUNT, etc.), GROUP BY clause for grouping results, HAVING clause for filtering grouped data	https://www.geeksforgeeks.org/mysql-aggregate-function/
		https://www.w3schools.com/mysql/mysql_groupby.asp
Joins and Subqueries	INNER JOIN, LEFT JOIN, RIGHT JOIN, Self-joins and cross joins, Subqueries in SELECT, WHERE, and FROM clauses, Correlated Subqueries	https://www.geeksforgeeks.org/mysql-join-1/
		https://www.geeksforgeeks.org/mysql-subquery/
Data Modification with INSERT, UPDATE, DELETE	Inserting data into tables, Updating existing records, Deleting records from tables	https://www.geeksforgeeks.org/sql-ddl-dql-dml-dcl-tcl-commands/
Creating and Modifying Tables	Creating tables with CREATE TABLE, Modifying table structure with ALTER TABLE, Deleting tables with DROP TABLE	https://www.geeksforgeeks.org/sql-create-table/?ref=lbp
		https://www.geeksforgeeks.org/sql-alter-add-drop-modify/?ref=lbp
Indexes and Constraints	Creating indexes for better query performance, Defining primary and foreign key constraints, Ensuring data integrity with UNIQUE and CHECK constraints	https://www.w3schools.com/mysql/mysql_constraints.asp
		https://www.w3schools.com/mysql/mysql_create_index.asp

Check Your Understanding:

[MySQL Quiz](#)

Python

Overview:

This module provides a comprehensive journey through the essential topics in python programming. Beginning with the setup of the Python development environment and basics of writing programs, Participants will grasp the fundamentals of Python syntax, data structures, functions, and modules. They will also explore advanced concepts such as object-oriented programming, file handling, and exception handling. Through hands-on coding exercises and projects, users will gain practical experience and the confidence to apply Python to various real-world scenarios.

Learning Objectives:

By the end of this module, learners will be able to:

- Understand the process of downloading and installing Python on Windows, Mac, and Linux.
- Learn various methods to run Python programs, including command line, IDLE, and script execution.
- Identify popular text editors and integrated development environments (IDEs) for Python development.
- Recognize and use Python's built-in data types, including strings, lists, tuples, dictionaries, and booleans.
- Understand what variables are and how to declare and use them in Python.
- Perform basic and advanced mathematical operations using Python's arithmetic operators.
- Capture and handle user input in Python programs.
- Implement control flow statements such as if, elif, and else to direct the execution of Python programs.
- Comprehend the significance of whitespace in Python syntax and structure.
- Organize Python code into functions and modules for better readability and reusability.
- Understand what modules are and how to use them in Python.
- Define and use functions to encapsulate code and promote reuse.
- Identify and use important built-in Python functions for common tasks.
- Read from and write to files using Python's file handling capabilities.
- Implement error handling in Python using try, except, and finally blocks.
- Utilize Python's collection objects, including lists, dictionaries, tuples, and sets.
- Understand the principles of object-oriented programming (OOP) in Python and create basic classes and objects.

- Design and implement custom classes in Python with attributes and methods.
- Apply the concepts of inheritance and polymorphism to create more flexible and reusable code.

Self-Learning (Open-source links):

Python		
Topics	Sub-Topics	Learning Reference Links
Python Basics	• Installing Python • Running Python Programs • Text Editors and IDEs	https://realpython.com/installing-python/ https://realpython.com/run-python-scripts/ https://realpython.com/python-ides-code-editors-guide/
Python Fundamentals	• Data Types • Variables • Mathematical Operations • User Input • Control Flow • Whitespace Importance • Organizing Programs	https://www.edureka.co/blog/variables-and-data-types-in-python/ https://www.geeksforgeeks.org/taking-input-in-python/ https://guicommits.com/organize-python-code-like-a-pro/
Advanced Python	• Modules • Functions • Built-in Functions • File I/O • Error Handling	https://www.geeksforgeeks.org/python-modules/ https://www.geeksforgeeks.org/python-functions/ https://www.w3schools.com/python/python_ref_functions.asp https://www.geeksforgeeks.org/file-handling-python/ https://www.geeksforgeeks.org/python-exception-handling/
Python Collections	• Lists, Dictionaries, Tuples, and Sets	https://www.pythonforall.com/python-collections https://www.tutorialsteacher.com/python/collections-module
Object-Oriented Python	• Classes • Custom Classes • Inheritance and Polymorphism	https://www.geeksforgeeks.org/python-classes-and-objects/ https://learnpython.com/blog/custom-class-python/ https://nsworldinfo.medium.com/understanding-inheritance-and-polymorphism-in-python-mastering-object-oriented-programming-part-b2b33168e963

Check Your Understanding:

[Python Quiz](#)

What Next

Congratulations on successfully completing the DN 5.0 Upskilling Learning Program!

As you have now finished this important phase of your learning, you will be taking a **Skill Based Assessment (SBA)** to certify your skills. **This assessment will cover all the skills and topics you have learned during Upskilling Learning Program**, ensuring you have a comprehensive understanding of the material.

We wish you the best of luck for your assessment and look forward to seeing you apply your newly acquired knowledge and skills. Good luck!